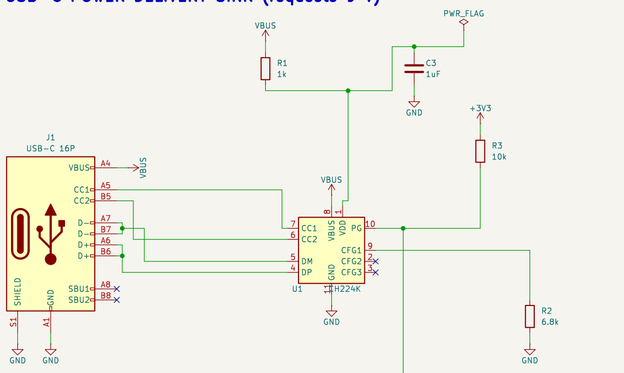
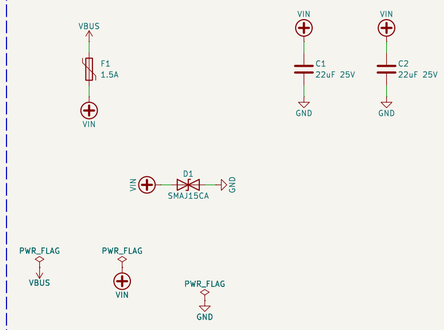
1 USB-C POWER DELIVERY SINK (requests 9 V)



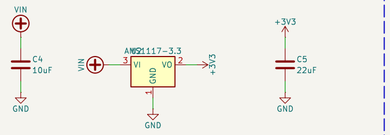
CFG1 resistor sets the PD request: 6.8k=9V 24k=12V 56k=15V open=20V

(firmware derates if only 5 V is granted, see VSENSE)

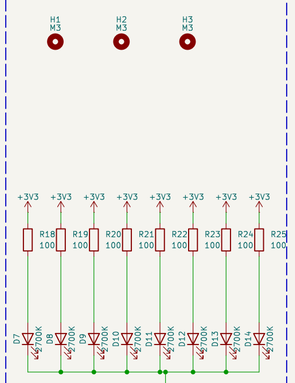
2 INPUT PROTECTION & BULK



3 3.3 V LOGIC SUPPLY

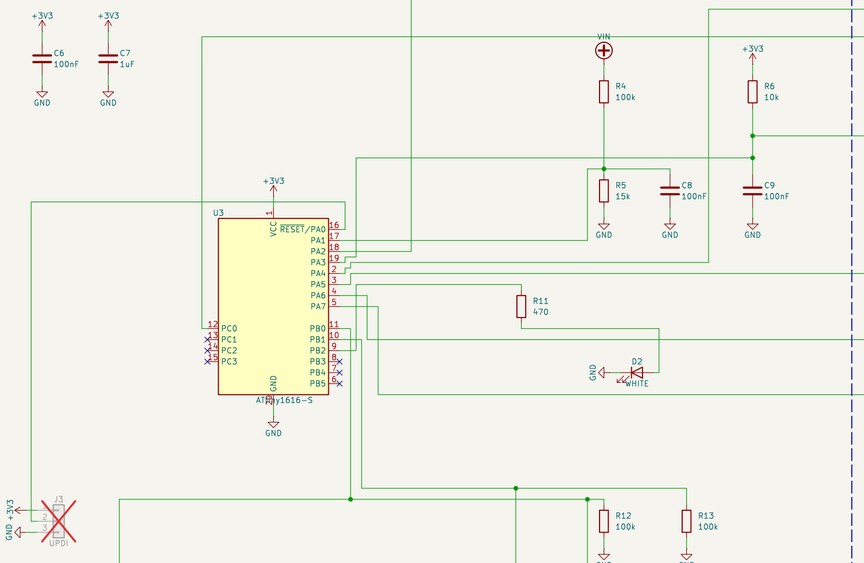


11 BODY GLOW (lights the opal base band) + MECHANICAL

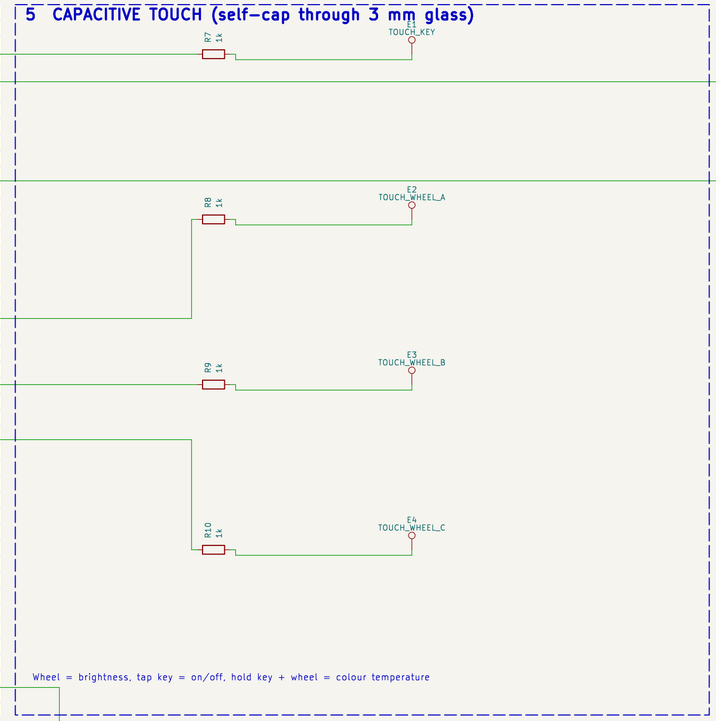


PCO = TCD0 WOC PWM: breathing night-glow through the base

4 MCU – ATtiny1616 (PTC touch, TCA0 PWM)

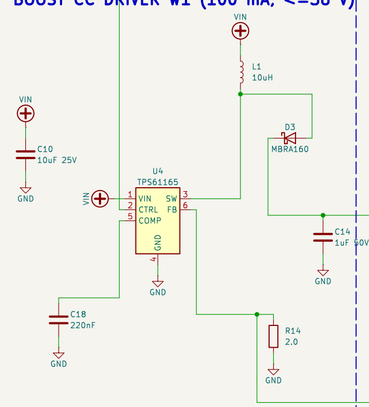


5 CAPACITIVE TOUCH (self-cap through 3 mm glass)

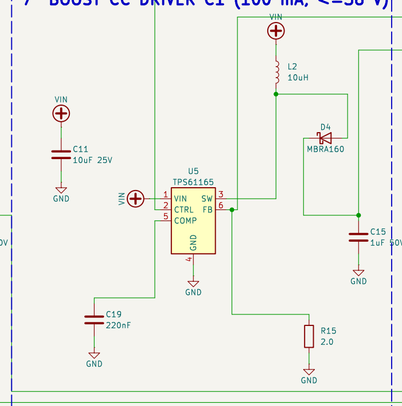


Wheel = brightness, tap key = on/off, hold key + wheel = colour temperature

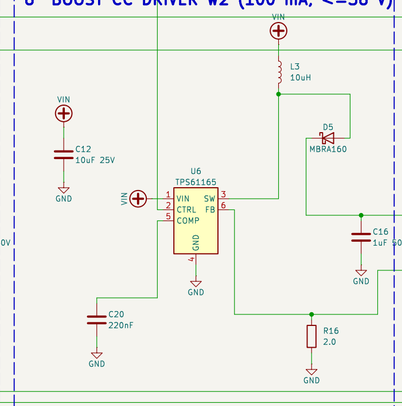
6 BOOST CC DRIVER W1 (100 mA, <=38 V)



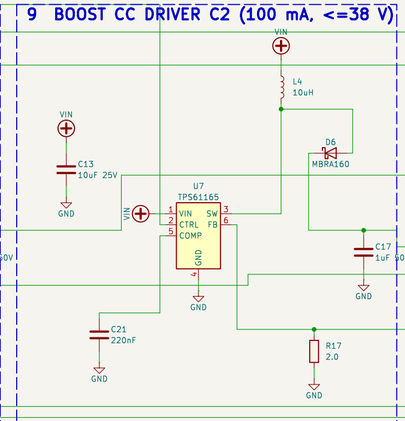
7 BOOST CC DRIVER C1 (100 mA, <=38 V)



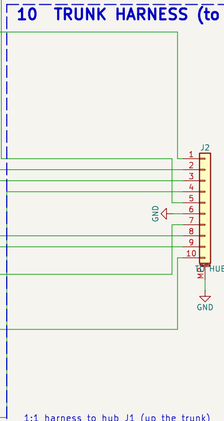
8 BOOST CC DRIVER W2 (100 mA, <=38 V)



9 BOOST CC DRIVER C2 (100 mA, <=38 V)



10 TRUNK HARNESS (to the tree hub)



1:1 harness to hub J1 (up the trunk)

J2: 10-wire harness up the trunk to the eclipse-hub board (1:1)
Board: 2 layers, 1.6 mm FR4, 1 oz, matte black mask, ENIG
USB-C PD 9 V sink, 4ch boost CC LED drivers, ATtiny1616 capacitive touch
ECLIPSE desk lamp – open hardware

Sheet: /	
File: eclipse-core.kicad_sch	
Title: ECLIPSE Tree – Core board (base): power, drivers, touch MCU	
Size: A2	Date: 2026-10-07
KiCad E.D.A. 9.0.9	Id: 1/1

Rev: A